Thesis Title

Thesis Subtitle



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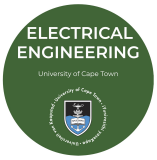
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Keywords: keyword1; keyword2; keyword3



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Mon 10 August, 2026

Abstract

Thesis Title

Joab Gray **Kloppers**

Mon 10 August, 2026

Acknowledgements

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List of Tables

1.1 Table title 2

List of Symbols

Symbol	Description	Unit
$\boldsymbol{B}$	Magnetic flux density	T
$\boldsymbol{E}$	Electric field	V/m
$\boldsymbol{J}$	Current density	A/m^2
t	Time	s
ε_0	Permittivity of free space (8.85×10^{-12})	$C^2/N{\cdot}m^2$
μ_0	Permeability of free space ($4\pi \times 10^{-7}$)	N/A^2

Chapter 1

Introduction

1.1 Background

1.1.1 Biological Activity

Sand prawn activity has a large effect on the surrounding ecosystem. Pillay and Branch [2] explain that by physically burrowing into the sediment, they distribute buried sediment to the surface, alter the surface structure via the introduced mounds, which can increase surface area, and alter both chemical and structural properties of the sediment through burrowing and nutrient mixing. Venter et al [3] further shows that sand prawn bi-directional water pumping filters can reduce phytoplankton biomass by up to 70% in estuarine environments which can help with ecosystem resilience.

1.1.2 Technical application

Schenone et al [1] demonstrates that surface microtopography can be used as a reliable metric to predict biochemical processes. However common metrics for measuring surface roughness characteristics such as (list them here) are often invasive and result in the damage of the very structures under measurement. Azhar et al [4] demonstrated that the use of low cost RGB-D cameras can be used to produce depth images of the sediment surface, which can then be processed to analyse the surface properties.

Sub-subsection

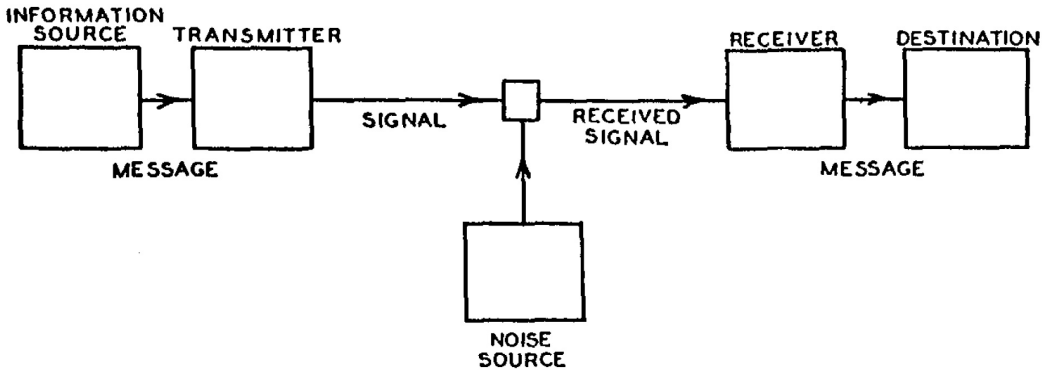


Figure 1.1: A simplified block diagram of a general communication system. A communication system can be abstracted as follows: an *information source* produces a message to be sent through a *transmitter* which converts it to a signal which can be sent via a communication *channel*. The received signal is noisy and is converted back to the message by the *receiver* which is then delivered to the *destination*. Taken from shannon1948

Table 1.1: Table title. Extended description of tables contents.

Heading 1	**Heading 2**	
	Subheading 1 [units]	*Subheading 2* [units]

1.2 Problem statement

Equations are numbered and include the chapter number. As they are considered part of the text, they include punctuation, and each parameter must be described in the text below. Ensure that the units are stated and where possible use SI units. For example, the Maxwell equations are expressed as:

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}, \tag{1.1}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1.2}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \tag{1.3}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}, \tag{1.4}$$

where $\boldsymbol{B}$ is the magnetic flux density in T, $\boldsymbol{J}$ is the total current density in A/m^2, $\boldsymbol{E}$ is the electric field in V/m, μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), and ϵ_0 is the permittivity of free space (8.85 $\times 10^{-12}$ C^2/N·m^2).

1.3 Project objectives

The Global Positioning System (GPS) is used for positioning. The Marginal Ice Zone (MIZ) is of particular interest in the Southern Ocean (SO).

1.4 Scope, limitations and assumptions

1.5 Plan of development

Chapter 2

Literature review

2.1 Biology

Pillay and Branch [2] describe how sand prawn species are prominent members in the benthic ecosystem. The sand prawns continuously change the sediment properties such as surface topography porosity and biochemistry, through actions such as burrowing and water pumping. These actions redistribute buried sediment to the surface which results in surface mounds which increase the effective surface area of the sediment by up to 15 times. These mounds can increase the effective hydrodynamic resistance which can help to stagnate the water flow close to the sediment. While the paper mentions measurements specific to sand prawn species located in other regions, they provide an insight into approximate feature sizes that can be expected, for example *Callianassa subterranea* present in the North sea can produce mounds 5.4cm tall and 11.7cm wide, and *Pestarella candida*, located in the Mediterranean and Adriatic Seas, which can produce mounds 3cm tall and 15cm wide.

Venter et al [3] further explains how sand prawns, *Callianassa kraussi*, present in the South African coastlines and at the Langebaan Lagoon, function as prominent water filterers. Within the burrows they actively irrigate the sediment through bi-directional pumping of water to replenish oxygen levels within the burrows, and to remove waste products. This irrigation has a secondary effect where phytoplankton are filtered out of the water by the burrow walls. This filtering can help improve ecosystem resilience to harmful conditions such as nutrient pollution which promote phytoplankton blooms. In relation to this project, the papers methodology entails manual visual counting of burrow openings.

The next question with regards to biological activity, is weather surface roughness, specifically rugosity, is a valid measurement to gauge the underlying biological activity within the benthic ecosystem. To this, Schenone et al [1, 5] determined experimentally that analysing two dimensional surface rugosity, they were able to correlate surface roughness measurements with sub-surface activity. The authors created a controlled laboratory environment to simulate the benthic ecosystem, which can be seen in Figure 2.1. The pipeline involved taking images perpendicular to the sediment cross section, manually extracting the surface line distance using the image processing application ImageJ, and computing the rugosity as a ration between this measured distance, and the distance of a straight line from end to end. Whilst the authors did find a correlation, rugosity explained 35% to 61% of variability observed, Their implementation is still under controlled laboratory conditions. The cross sectional profile imaging is not possible in situ, and the two-dimensional rugosity calculation relies on a thin 'slice' of sediment. These limitations point to a need for three dimensional surface scans.

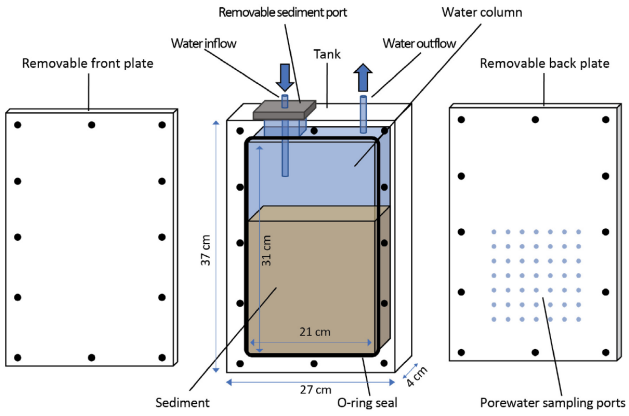


Figure 2.1: Custom incubation and flow through chamber by Schenone et al [1]

Chapter 3

Theoretical development

Chapter 4

Design description

Chapter 5

Experimental design

Chapter 6

Results

Chapter 7

Discussion

Chapter 8

Conclusions

Chapter 9

Further work

Appendix A

GitHub repository

This appendix provides a link to the GitHub repository where the code and additional materials for this final year project report can be found.

A.1 Title for appendix A.1

- GitHub Repository: Repo name

Appendix B: Ethics approval

Appendix C: Topic description

The original topic description used during the project selection process is included below. This topic has since been further refined, and the exact problem that was investigated is described in Section 1.2.

Appendix D: ECSA Graduate Attributes

Project mapping to ECSA Graduate Attributes

The table below provides a structured reflection on where each of the required Graduate Attributes (GAs) as defined by Engineering Council of South Africa (ECSA) associated with the final year project is met. This includes reference to the specific Chapter/Section (where applicable).

GA	Graduate Attribute Description	Student reflection: How this outcome was achieved during the project
GA1	*Problem solving*	
GA4	*Investigations, experiments and data analysis*	
GA5	*Use of engineering tools*	
GA6	*Professional and technical communication*	
GA8	*Individual and collaborative teamwork*	
GA9	*Independent learning ability*	

Bibliography

[1] S. Schenone, I. Bartl, and S. F. Thrush, "Sediment microtopography predicts localised benthic ecosystem functioning and the effect of species interactions," *Marine Ecology Progress Series*, vol. 749, pp. 47–55, Nov. 2024.

[2] D. Pillay and G. Branch, "Bioengineering effects of burrowing thalassinidean shrimps on marine soft-bottom ecosystems," in *Oceanography and Marine Biology: An Annual Review*, Jun. 2011, vol. 49, pp. 137–192.

[3] O. Venter, D. Pillay, and K. Prayag, "Water filtration by burrowing sandprawns provides novel insights on endobenthic engineering and solutions for eutrophication," *Scientific Reports*, vol. 10, no. 1, p. 1913, Feb. 2020.

[4] M. Azhar, J. R. Hillman, T. Gee, S. Schenone, W. van der Mark, S. F. Thrush, and P. Delmas, "An RGB-D framework for capturing soft-sediment microtopography," *Methods in Ecology and Evolution*, vol. 13, no. 8, pp. 1730–1745, 2022.

[5] S. Schenone, J. E. Hewitt, J. Hillman, R. Gladstone-Gallagher, J. Gammal, C. Pilditch, A. M. Lohrer, E. Ferretti, M. Azhar, P. Delmas, and S. F. Thrush, "Seafloor sediment microtopography as a surrogate for biodiversity and ecosystem functioning," *Ecological Applications*, vol. 35, no. 1, p. e3069, 2025.